

Matrices, Transformations & Eigenstructure

Mathematics Foundations — Session 2

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Today's Goal

- See a matrix not just as a grid of numbers, but as something that acts on space
 - Multiply matrices by hand and trace a vector through the forward pass of a small network
 - Read eigenvectors as the directions a transformation leaves pointing the same way, with PCA as the ML use case
- Deliverable: a by-hand two-layer forward pass, plus the max-variance eigen-direction on Iris

Plan

1. Opening Quiz
2. Matrix Definitions & Base Operations
3. Linear Transformations
4. Eigenvalues & Eigenvectors
5. Machine Learning
6. Recap
7. To Do for Next Session

Opening Quiz

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Opening Quiz: Vectors & the Dot Product

Matrix Definitions & Base Operations

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What Are Matrices?

Matrices: Example and Elements

Matrices in Machine Learning

Matrix Properties

Matrix–Vector and Matrix–Matrix Multiplication

Is $AB = BA$?

Multiplying Matrices of Different Sizes

Special Matrices

Vocabulary

Exercises (1/2)

Exercises (2/2)

Linear Transformations

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Introduction

What Is a Linear Transformation?

Linear Transformations and Geometry

The Determinant

Rank

Eigenvalues & Eigenvectors

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Eigenvalues

Eigenvectors

Machine Learning

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The Forward Pass

Deliverable

A by-hand two-layer forward pass, plus naming the eigen-direction of maximum variance on a small dataset (Iris).

Recap

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Recap

To Do for Next Session

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To Do for Next Session
